

Groups and Lie Algebras: Exercise Sheet 7

1. Der A is a Lie subalgebra

Consider a finite dimensional associative algebra A . A linear map $D \in \text{End } A$ is a derivation of A if

$$D(x \cdot y) = D(x) \cdot y + x \cdot D(y) .$$

To $\text{End } A$ we associate a Lie algebra $\mathfrak{gl}(A)$, where the bracket is the commutator. Show that $\text{Der}(A)$, (the space of derivations), is a subalgebra of $\mathfrak{gl}(A)$, where the bracket is again given by the commutator. What is the corresponding Lie group with Lie algebra $\text{Der}(A)$?

2. Representations of Lie algebras

Let $\mathfrak{g}$ be a Lie algebra and V, W representation spaces. Show that

- a) there is a representation on the direct sum $V \oplus W$ of vector spaces given by

$$((R_V \oplus R_W)(x))(v \oplus w) := R_V(x)v \oplus R_W(x)w, \quad \forall x \in \mathfrak{g}, v \in V, w \in W$$

- b) there is a representation on the tensor product $V \otimes W$ of vector spaces given by

$$((R_V \otimes R_W)(x))(v \otimes w) := R_V(x)v \otimes w + v \otimes (R_W(x)w), \quad \forall x \in \mathfrak{g}, v \in V, w \in W$$

- c) the following does *not* define a representation on $V \otimes W$

$$((R_V \otimes R_W)(x))(v \otimes w) := R_V(x)v \otimes R_W(x)w$$

- d) Let G be a Lie group with Lie algebra $\mathfrak{g}$. If the representations V and W are induced by representations of G on V and W , then the definitions above are compatible with the definition of $\oplus$ and $\otimes$ for G -representations from the lecture.

3. Angular momentum operators in quantum mechanics

- a) Let $\mathfrak{g}$ be a real Lie algebra. The *complexification* of $\mathfrak{g}$ is the complex Lie algebra

$$\mathfrak{g}_{\mathbb{C}} := \mathfrak{g} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathfrak{g} \oplus i\mathfrak{g},$$

with the $\mathbb{C}$ -bilinear extension of the Lie bracket defined by

$$[x \otimes z, y \otimes w] := [x, y] \otimes zw, \quad x, y \in \mathfrak{g}, \quad z, w \in \mathbb{C}.$$

The original algebra $\mathfrak{g}$ embeds into $\mathfrak{g}_{\mathbb{C}}$ as a real Lie subalgebra via $x \mapsto x \otimes 1$. Compute the Lie algebra of the group $SL(n)$ of complex matrices with determinant 1 and show that it is isomorphic to the complexification of $\mathfrak{su}(n)$ the Lie algebra of $SU(n)$.

- b) Consider the Hilbert space $L^2(\mathbb{R}^3)$ with the action of the usual position operators x_i and momentum operators $p_i = -i\frac{\partial}{\partial x_i}$ for $i = 1, 2, 3$. Show that the angular momentum operators $L_i = \sum_{j,k} \epsilon_{ijk} x_j p_k$ (the components of $L = x \times p$) form a representation of the complex Lie algebra $\mathfrak{sl}(2, \mathbb{C})$.
- c) Describe the decomposition into irreducible representations using the spherical harmonics from the previous exercise sheet.